



Classification on Breast Cancer Data

Presented

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Abstract

In this project, we carry out a detailed statistical analysis on the Wisconsin Breast Cancer dataset to develop a reliable model for classifying tumors as benign or malignant. The initial phase of the study involves thorough data preprocessing, including the detection and removal of multicollinearity among predictor variables to improve model interpretability and performance. We also assess key distributional assumptions by conducting tests for multivariate normality based on Mahalanobis distances, which guide the choice of appropriate modeling techniques. Following this, several classical statistical classification models are applied to the data, and their performances are systematically evaluated using a variety of metrics such as accuracy, precision, recall, F1-score, confusion matrices, and receiver operating characteristic (ROC) curves. To ensure model robustness and generalizability, cross-validation techniques are employed, and the area under the ROC curve (AUC) is used as a key criterion for comparison. The comparative analysis reveals consistently high performance across models, with one showing slightly better generalization capabilities. This project underscores the effectiveness of traditional statistical approaches in handling real-world biomedical classification problems and emphasizes the importance of assumption checking, diagnostic testing, and comparative evaluation in model selection.

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Chapter 1

Introduction

Breast cancer is one of the most prevalent and life-threatening diseases that affect women worldwide. Early and accurate diagnosis plays a critical role in improving patient outcomes and survival rates. In recent years, statistical and machine learning methods have gained prominence for their ability to classify medical data effectively. This project focuses on applying classification techniques to breast cancer data to distinguish between malignant and benign tumors. Using a combination of exploratory data analysis and supervised learning algorithms, our objective is to evaluate the performance of models such as Linear Discriminant Analysis and others in accurately predicting tumor classes based on relevant clinical features.

1.1 Looking at the data

The Wisconsin Breast Cancer dataset is one of the most widely used benchmark datasets for evaluating classification algorithms in the field of medical diagnostics. Originally curated by Dr. William H. Wolberg of the University of Wisconsin Hospitals, Madison, it contains features computed from digitized images of fine needle aspirate (FNA) of breast masses. Each instance is labeled as either benign or malignant, making it highly suitable for binary classification tasks. Due to its balanced class distribution, well-structured features, and real-world relevance, the dataset has become a standard in machine learning research and educational projects. The dataset is publicly available from the UCI Machine Learning Repository at the following link: [https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+\(Diagnostic\)](https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)). The data has more than 30 columns, i.e. more than 30 variables of observation for each patient and the complete data of more than 550 patients. Here I present few columns and rows for a glimpse of how the data looks. The 1 in the diagnosis column means malignant and 0 means benign.

Index	Diagnosis	Radius Mean	Texture Mean	Area Mean	Smoothness Mean
0	1	17.99	10.38	1001.0	0.11840
1	1	20.57	17.77	1326.0	0.08474
2	1	19.69	21.25	1203.0	0.10960
3	1	11.42	20.38	386.1	0.14250
4	1	20.29	14.34	1297.0	0.10030
5	1	12.45	15.70	477.1	0.12780
6	1	18.25	19.98	1040.0	0.09463
7	1	13.71	20.83	577.9	0.11890
8	1	13.00	21.82	519.8	0.12730
9	1	12.46	24.04	475.9	0.11860

Table 1.1: Sample of the breast cancer dataset (first 10 rows)

Chapter 2

Preprocessing and Analysis of the Data

Before applying any classification algorithms, it is essential to preprocess and analyze the dataset to ensure the models receive clean, consistent, and meaningful input. Real-world data often contains noise, missing values, redundant features, or variables with different scales—all of which can adversely affect model performance. Through preprocessing, we handle such issues by cleaning the data, selecting relevant features, and standardizing inputs. Additionally, exploratory data analysis helps us understand the underlying structure, relationships, and distributions within the dataset, guiding informed decisions for model building and improving overall interpretability.

2.1 Checking Class Distribution

Before applying classification algorithms, it is crucial to examine the distribution of the target classes in the dataset. An imbalanced class distribution—where one class significantly outnumbers the other—can lead to biased models that favor the majority class. This often results in high overall accuracy but poor performance in detecting the minority class, which, in medical applications like breast cancer detection, can be critical. For example, failing to correctly identify malignant cases due to class imbalance could have serious real-world consequences. By analyzing class distribution early, we can decide whether techniques like resampling (oversampling or undersampling), synthetic data generation (e.g., SMOTE), or using metrics beyond accuracy (such as precision, recall, or F1-score) are necessary to ensure fair and effective classification. In our case, both classes—benign and malignant—have a sufficient number of instances, so we can proceed without the need for special handling of class imbalance.

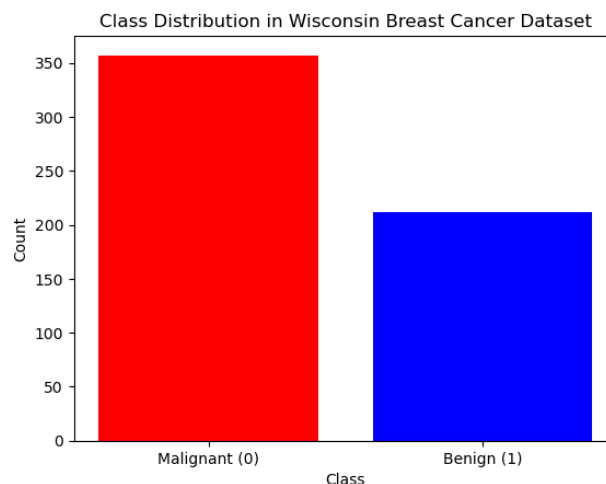


Figure 2.1: Distribution of instances between the Two Classes

2.2 Checking for Multicollinearity

Multicollinearity refers to a situation where two or more predictor variables in a dataset are highly linearly correlated. This can pose significant problems for many classification algorithms, especially those that rely on linear decision boundaries, such as Linear Discriminant Analysis (LDA). When predictors are highly correlated, it becomes difficult to estimate the individual effect of each variable on the target outcome, as changes in one variable are associated with changes in another. This can lead to unstable model coefficients, inflated variances, and poor interpretability of the model. Moreover, multicollinearity can reduce the predictive power and generalizability of the model, particularly when applied to new data. Therefore, identifying and addressing multicollinearity—through techniques like correlation analysis—is a crucial step in building robust and reliable classification models.

To assess multicollinearity in our dataset, we computed the correlation matrix and visualized it using a heatmap. This helped us identify pairs of features that are highly correlated with each other. In particular, we focused on variables that had a correlation coefficient greater than 0.9 with any other variable, as such strong linear relationships can introduce redundancy and instability in classification models. To mitigate this issue, we removed one variable from each highly correlated pair, thereby reducing multicollinearity while preserving the overall information content of the dataset. This step not only improved model interpretability but also contributed to better generalization. The correlation heatmap before variable removal is shown below for reference.

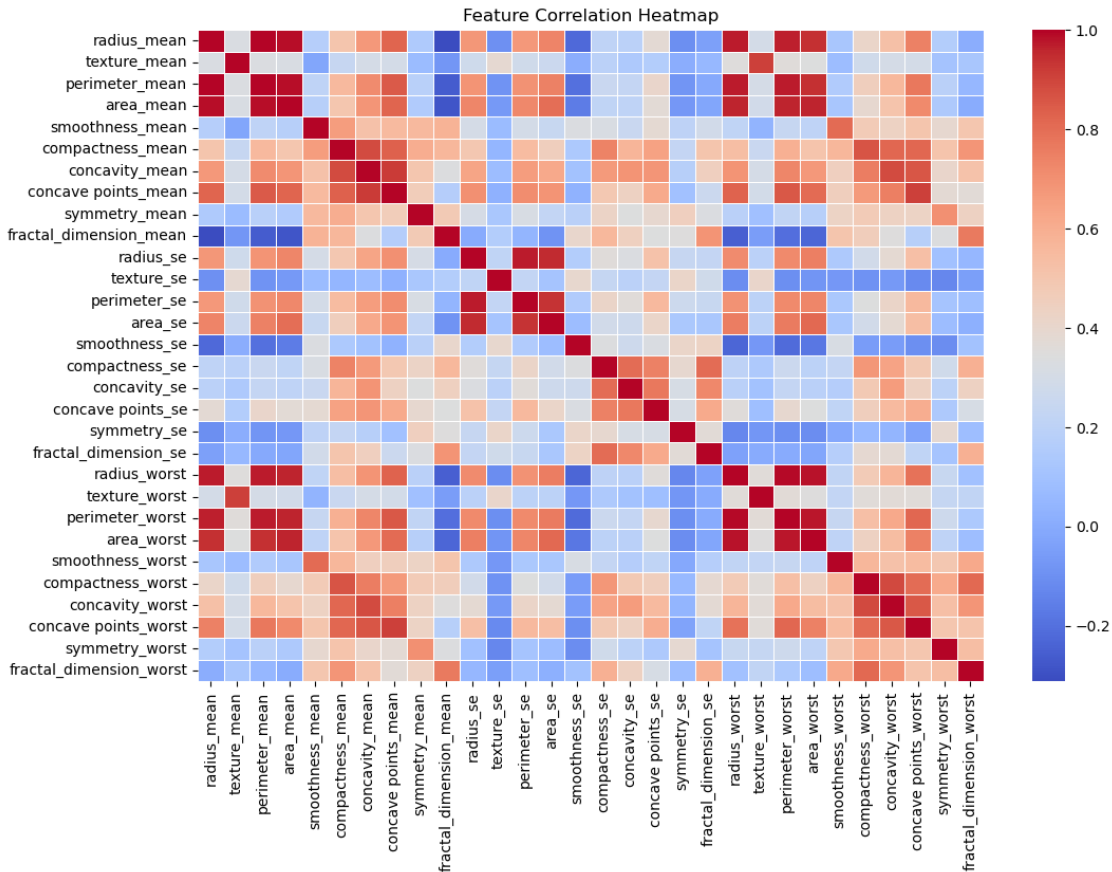


Figure 2.2: Correlation heatmap of the features before removing highly correlated variables.

Features removed due to high correlation are listed here 'perimeter-mean', 'area-mean', 'concave points-mean', 'perimeter-se', 'area-se', 'radius-worst', 'texture-worst', 'perimeter-worst', 'area-worst', 'concave points-worst'. Here, onward, we shall work with the trimmed data only.

2.3 Checking Normality of Variables

In this project, assessing the normality of variables is particularly important because some classification algorithms, such as Linear Discriminant Analysis (LDA), assume that the input features are normally distributed within each class. The performance and theoretical validity of these models depend on this assumption. If the variables deviate significantly from normality—exhibiting skewness or heavy tails—the estimated parameters may be biased, and the model’s classification accuracy can deteriorate. Additionally, checking for normality helps inform preprocessing decisions such as applying transformations (e.g., logarithmic or Box-Cox) to make the data more suitable for the chosen algorithm. Even in cases where algorithms are robust to non-normal data, understanding the distributional shape of the variables aids in better interpretation, visualization, and feature engineering.

To gain an initial understanding of the distribution of the variables, we begin by examining Q-Q (quantile-quantile) plots for a few selected features. Q-Q plots provide a visual comparison between the quantiles of a variable and those of a normal distribution, helping us detect deviations from normality such as skewness or heavy tails. This crude yet effective method gives us a quick insight into whether the normality assumption holds reasonably well for the data. For illustration, we have included two representative Q-Q plots at the end of this paragraph.

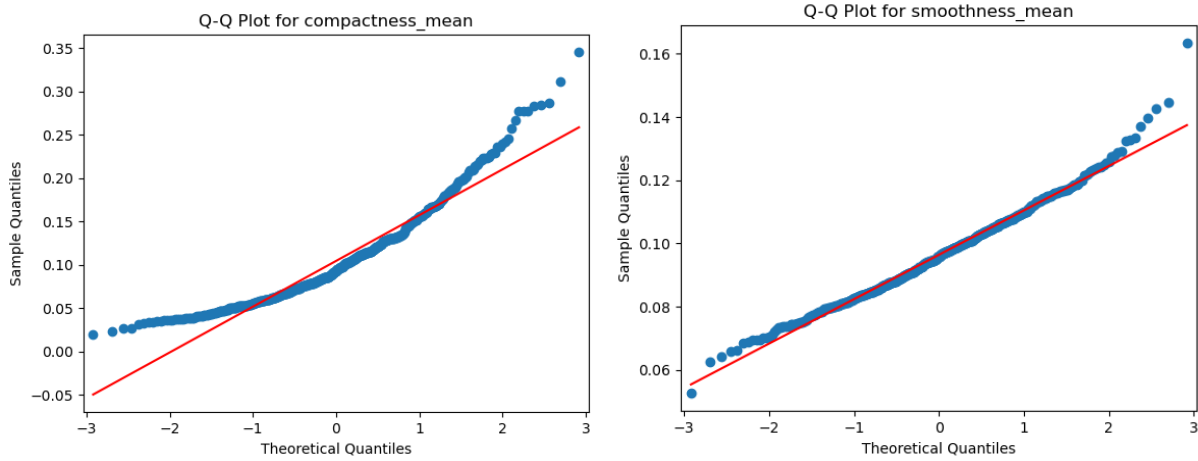


Figure 2.3: Q-Q plots for the variables `area_mean` and `smoothness_mean`.

We can observe that the variable smoothness-mean has a Q-Q plot clearly indicating the underlying normality, but that is not the case for the variable compactness-mean. The Q-Q plot for compactness-mean clearly indicates that the variable is not likely to follow the normality assumption. Here the two polar opposite examples were picked just to highlight the different behavior of variables in the data, so to make a concrete conclusion, we take the following approach.

2.3.1 The Kolmogorov Smirnov Test for Multivariate Normality

When testing for multivariate normality, a common approach is to compute the **Mahalanobis distance** of each data point from the mean of the multivariate distribution. The Mahalanobis distance accounts for correlations between variables and is given by the formula:

$$D_M(x) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)},$$

where: - x is the vector of the data point, - μ is the vector of the mean of the multivariate distribution, - Σ is the covariance matrix of the data.

The Mahalanobis distance measures the number of standard deviations a point is away from the mean, considering the covariance structure of the data. For a multivariate normal distribution, the squared Mahalanobis distance should follow a chi-squared (χ^2) distribution with degrees of freedom equal to the number of variables.

To assess the normality of the data, we can use the **Kolmogorov–Smirnov (K-S) test** on the Mahalanobis distances. Specifically, the following steps are performed:

1. Compute the Mahalanobis distance for each data point.
2. Standardize the Mahalanobis distances by comparing them to the expected χ^2 distribution.
3. Perform the Kolmogorov–Smirnov test on the empirical distribution of the standardized Mahalanobis distances against the χ^2 distribution.

Under the null hypothesis of the Kolmogorov–Smirnov test the Mahalanobis distances follow a χ^2 distribution. If the test statistic is large (i.e., the p-value is small), we reject the null hypothesis, suggesting that the data do not follow a multivariate normal distribution.

This method of testing for multivariate normality is particularly useful when dealing with multiple correlated variables, as it incorporates the correlation structure between the features in the dataset, unlike univariate tests that consider each variable independently.

Here in our data, it is enough to test if the data is distributed as a multivariate normal, we don't need every variable to be normal independently, so we perform this Multivariate Kolmogorov–Smirnov test on our data.

To assess multivariate normality, we applied the Kolmogorov–Smirnov (K–S) test on the Mahalanobis distances computed from the dataset. The test compared the empirical distribution of these distances to the theoretical chi-squared distribution with degrees of freedom equal to the number of variables.

The resulting test statistic and p-value were examined to draw a conclusion. In our case, the p-value was found to be **7.881185149940329e-89**, which is **less** than the commonly used significance level of 0.05. Therefore, we **reject** the null hypothesis that the data follows a multivariate normal distribution.

This result provides **evidence** that the dataset **deviates from** the assumption of multivariate normality, which is an important consideration for subsequent modeling steps such as Linear Discriminant Analysis (LDA) that assume normality of the features.

Chapter 3

Applying Classification Algorithms

In this section, we move on to applying classification algorithms to the breast cancer dataset. Initially, the focus was on exploring methods such as Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA), which are well-known for their interpretability and performance under certain assumptions. However, both LDA and QDA rely heavily on the assumption that the predictor variables follow a multivariate normal distribution.

To validate this assumption, we previously conducted the Kolmogorov–Smirnov test on the Mahalanobis distances of the data. The results indicated a statistically significant deviation from multivariate normality. Given this, applying LDA would not be logically justified, as it assumes homoscedasticity and normality of the predictors. Therefore, we choose not to proceed with LDA in this analysis. Instead, we shift our focus toward classification algorithms that are more robust to deviations from normality, such as QDA, logistic regression, and other methods.

In this chapter, we apply four different classification algorithms to the breast cancer dataset in order to evaluate their performance and effectiveness. The methods under consideration are **Fisher Discriminant Analysis (FDA)**, **Quadratic Discriminant Analysis (QDA)**, **Logistic Regression**, and **Probit Regression**. These techniques are widely used in statistical classification problems and provide a good balance between interpretability and predictive power. Each method will be trained and tested on the dataset, and their classification capabilities will be compared using standard performance metrics such as accuracy and the ROC curve.

3.1 Fisher Discriminant Analysis

Fisher’s Discriminant Analysis (FDA) is a classification method that projects high-dimensional data onto a lower-dimensional space—specifically, a line—such that the separation between different classes is maximized. The technique works by finding a linear combination of features that best separates the classes based on two criteria: maximizing the between-class variance while minimizing the within-class variance. Mathematically, it involves solving an optimization problem where the optimal projection vector is obtained by maximizing the Fisher criterion, which is the ratio of the determinant of the between-class scatter matrix to that of the within-class scatter matrix. Once the projection is done, classification is performed by modeling the class-conditional distributions as Gaussians and applying Bayes’ theorem. FDA assumes that the different classes share the same covariance matrix, which results in a linear decision boundary. But we know for a fact that the Linear Discriminant Analysis gives the same results as the Fisher Discriminant Analysis. We’ll apply that result here.

Performance of Fisher Discriminant Analysis

We applied **Fisher’s Discriminant Analysis (FDA)** to the breast cancer dataset after performing an 80-20 train-test split. The model achieved a high **accuracy of 95.6%** on the test data. The cross-validation score for FDA is **0.9561 ± 0.0251** , indicating a stable and reliable model performance across different data splits. A detailed classification report is presented in Table 3.1, and the confusion matrix is shown in Table 3.2.

From the classification report, we observe that the model performs very well across both classes. For the benign class (label 0), the precision is 0.95, recall is 0.99, and F1-score is 0.97, indicating that almost all benign cases were correctly identified with very few false positives. For the malignant class (label 1), the precision is even higher at 0.97, although the recall drops to 0.91, meaning that while the predictions for malignant cases are reliable, a few malignant cases were misclassified as benign. The F1-score of 0.94 for class 1 reflects this balance.

The macro average F1-score of 0.95 shows the model performs well on both classes without being biased toward one class due to imbalance, and the weighted average F1-score of 0.96 confirms this consistency across all instances.

The confusion matrix in Table 3.2 further supports these observations: out of 71 benign samples, 70 were correctly classified and 1 was misclassified as malignant. Out of 43 malignant samples, 39 were correctly classified and 4 were misclassified as benign. This high level of precision and recall makes FDA a strong candidate for classification in this context.

Table 3.1: Classification Report for Fisher Discriminant Analysis

Class	Precision	Recall	F1-score	Support
0 (Benign)	0.95	0.99	0.97	71
1 (Malignant)	0.97	0.91	0.94	43
Accuracy	0.956			
Macro Avg	0.96	0.95	0.95	114
Weighted Avg	0.96	0.96	0.96	114
Cross Validation Score	0.9561 $\pm$ 0.0251			

Table 3.2: Confusion Matrix for Fisher Discriminant Analysis

Actual \ Predicted	0 (Benign)	1 (Malignant)
0 (Benign)	70	1
1 (Malignant)	4	39

3.2 Quadratic Discriminant Analysis

Quadratic Discriminant Analysis (QDA) is a classification technique based on Bayes' theorem that assumes each class follows a multivariate normal distribution, but unlike Linear Discriminant Analysis (LDA), QDA allows each class to have its own distinct covariance matrix. This results in quadratic decision boundaries, providing greater flexibility in modeling the relationship between the features and the target class.

Mathematically, QDA models the probability density function of each class k as:

$$P(x | y = k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right)$$

where μ_k and Σ_k are the mean vector and covariance matrix of class k , respectively, and d is the number of features. The classifier assigns a new observation x to the class with the highest posterior probability $P(y = k | x)$, computed using Bayes' theorem.

QDA is especially powerful when the class distributions exhibit differing variances or correlations among variables. While this flexibility can lead to better classification performance in such cases, it also increases the risk of overfitting when the number of features is large compared to the number of observations.

Although **Quadratic Discriminant Analysis (QDA)** assumes that the data from each class follows a multivariate normal distribution, it has been shown to be relatively robust to deviations from this assumption. In practice, QDA often performs well even when the underlying data is

not strictly normal. This is because QDA does not rely solely on the normality assumption for classification; it instead utilizes the estimated class-specific mean vectors and covariance matrices to construct decision boundaries that separate the classes effectively.

Moreover, QDA’s flexibility in modeling different covariance structures for each class allows it to adapt better to real-world data where such structures naturally vary. While extreme departures from normality or the presence of outliers can affect performance, moderate non-normality does not necessarily invalidate the use of QDA. Therefore, despite our earlier analysis indicating that the data is not perfectly normally distributed (as revealed by the Kolmogorov-Smirnov test on Mahalanobis distances), we proceed with QDA under the understanding that it can still yield meaningful and accurate classification results in such scenarios.

Performance of Quadratic Discriminant Analysis

We applied **Quadratic Discriminant Analysis (QDA)** to the breast cancer dataset following the same train-test split and preprocessing as in the previous case. The model achieved an overall **accuracy of 93.86%** on the test data. The cross-validation score for QDA is **0.9543 ± 0.0251** , indicating that it performs similarly to FDA with stable accuracy across data splits. A detailed summary of the performance metrics is shown in Table 3.3, and the corresponding confusion matrix is provided in Table 3.4.

From the classification report, we observe strong predictive performance across both classes. For class 0 (Benign), the precision is 0.94, recall is 0.96, and F1-score is 0.95, indicating the model’s high ability to identify benign cases correctly. For class 1 (Malignant), the precision is 0.93 and recall is 0.91, with an F1-score of 0.92 — reflecting reliable detection of malignant tumors, although a few are misclassified as benign.

The macro average F1-score is 0.93, showing balanced performance across both classes, while the weighted average F1-score of 0.94 confirms strong overall accuracy. The confusion matrix in Table 3.4 shows that 68 out of 71 benign cases and 39 out of 43 malignant cases were correctly classified. These results demonstrate that QDA performs effectively on this dataset, even with some deviation from the normality assumption.

Table 3.3: Classification Report for Quadratic Discriminant Analysis

Class	Precision	Recall	F1-score	Support
0 (Benign)	0.94	0.96	0.95	71
1 (Malignant)	0.93	0.91	0.92	43
Accuracy	0.9386			
Macro Avg	0.94	0.93	0.93	114
Weighted Avg	0.94	0.94	0.94	114
Cross Validation Score	0.9543 ± 0.0251			

Table 3.4: Confusion Matrix for Quadratic Discriminant Analysis

Actual \ Predicted	0 (Benign)	1 (Malignant)
0 (Benign)	68	3
1 (Malignant)	4	39

3.3 Logistic Regression

Logistic Regression is a statistical model commonly used for binary classification tasks, where the objective is to predict the probability of an outcome belonging to one of two classes. Unlike linear regression, which predicts continuous values, logistic regression outputs a probability between 0 and 1, making it suitable for classification problems.

The model uses the **logistic function** (also known as the sigmoid function) to map the linear combination of input features into a probability. The logistic function is defined as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

where:

- $P(Y = 1|X)$ is the probability of the positive class (class 1),
- $X_1, X_2, \dots, X_n$ are the input features,
- $\beta_0, \beta_1, \dots, \beta_n$ are the model coefficients (parameters) that are learned during the training process.

Logistic regression uses **maximum likelihood estimation** (MLE) to find the optimal values for these coefficients. After training, the model classifies an observation as belonging to class 1 if the predicted probability is greater than or equal to a threshold (commonly 0.5). Otherwise, it is classified as class 0.

Logistic regression is a widely used, efficient, and interpretable method, particularly when dealing with linear decision boundaries. However, it assumes that the relationship between the features and the log-odds of the outcome is linear, and it performs best when the data is linearly separable.

Logistic Regression Results

We applied **Logistic Regression** to the breast cancer dataset after similar preprocessing as previous cases. The model achieved a high **accuracy of 95.6%** on the test data. The cross-validation score for Logistic Regression is **0.9332 ± 0.0313**, indicating stable and reliable model performance across different data splits. A detailed classification report is presented in Table 3.5, and the confusion matrix is shown in Table 3.6.

From the classification report, we observe that the model performs well across both classes. For the benign class (label 0), the precision is 0.96, recall is 0.97, and F1-score is 0.97, indicating that almost all benign cases were correctly identified with very few false positives. For the malignant class (label 1), the precision is 0.95, while the recall is 0.93, meaning that the model correctly identified most of the malignant cases but misclassified a few as benign. The F1-score of 0.94 for class 1 reflects this balance.

The macro average F1-score of 0.95 shows the model performs well on both classes, and the weighted average F1-score of 0.96 confirms the model's consistency across all instances.

The confusion matrix in Table 3.6 further supports these observations: out of 71 benign samples, 69 were correctly classified and 2 were misclassified as malignant. Out of 43 malignant samples, 40 were correctly classified and 3 were misclassified as benign. This high level of precision and recall makes Logistic Regression a reliable candidate for classification in this context.

Table 3.5: Classification Report for Logistic Regression

Class	Precision	Recall	F1-score	Support
0 (Benign)	0.96	0.97	0.97	71
1 (Malignant)	0.95	0.93	0.94	43
Accuracy	0.956			
Macro Avg	0.96	0.95	0.95	114
Weighted Avg	0.96	0.96	0.96	114
Cross Validation Score	0.9332 ± 0.0313			

Table 3.6: Confusion Matrix for Logistic Regression

Actual \ Predicted	0 (Benign)	1 (Malignant)
0 (Benign)	69	2
1 (Malignant)	3	40

3.4 Probit Regression

Probit Regression is a type of regression used for modeling binary outcome variables. It is similar to logistic regression but uses the cumulative distribution function (CDF) of the standard normal distribution instead of the logistic function. In probit regression, the probability of the binary response variable being 1 is modeled using the **probit function**, which is the inverse of the normal CDF, often referred to as the *probit link function*. The relationship between the predictor variables and the binary outcome is expressed as:

$$P(Y = 1|X) = \Phi(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)$$

where Φ represents the CDF of the standard normal distribution, $\beta_0, \beta_1, \dots, \beta_p$ are the coefficients to be estimated, and $X_1, X_2, \dots, X_p$ are the predictor variables. The model assumes that the latent variable, representing the underlying propensity of an event occurring, follows a normal distribution. Probit regression is particularly useful when the data is believed to follow a normal distribution but logistic regression may not be as appropriate. It provides a direct way to interpret the probability of success based on the normal distribution, and it can be seen as an alternative to logistic regression when modeling binary outcomes.

Performance of Probit Regression

We applied **Probit Regression** to the breast cancer dataset without standardizing the features and using an 80-20 train-test split. The model achieved a test accuracy of **95.6%**. Additionally, we manually performed 10-fold cross-validation and obtained a cross-validation score of **0.9561 $\pm$ 0.0238**, indicating high model stability and generalization ability. The classification report and confusion matrix are presented in Table 3.7 and Table 3.8, respectively.

From the classification report, we observe that the model exhibits strong performance across both classes. For the benign class (label 0), the precision is 0.97, recall is 0.96, and F1-score is 0.96, indicating a high proportion of true benign identifications with minimal false positives. For the malignant class (label 1), the precision is 0.93, recall is 0.95, and F1-score is 0.94, showing reliable identification of malignant cases with few false negatives.

The macro average and weighted average F1-scores are both around 0.95–0.96, suggesting that the model performs consistently across both classes, even with class imbalance.

The confusion matrix in Table 3.8 further supports these findings: 68 out of 71 benign samples were correctly classified, with 3 misclassified as malignant. Similarly, 41 out of 43 malignant cases were correctly predicted, with only 2 benign misclassifications. This balanced performance confirms the suitability of probit regression for this classification problem.

Table 3.7: Classification Report for Probit Regression

Class	Precision	Recall	F1-score	Support
0 (Benign)	0.97	0.96	0.96	71
1 (Malignant)	0.93	0.95	0.94	43
Accuracy	0.956			
Macro Avg	0.95	0.96	0.95	114
Weighted Avg	0.96	0.96	0.96	114
Cross Validation Score	0.9561 $\pm$ 0.0238			

Table 3.8: Confusion Matrix for Probit Regression		
Actual \ Predicted	0 (Benign)	1 (Malignant)
0 (Benign)	68	3
1 (Malignant)	2	41

Conclusion

In this chapter, we applied four distinct classification techniques on the breast cancer dataset—**Fisher’s Linear Discriminant Analysis (FDA)**, **Quadratic Discriminant Analysis (QDA)**, **Logistic Regression**, and **Probit Regression**. Each model was evaluated using several performance metrics, including accuracy, precision, recall, F1-score, confusion matrix, and cross-validation scores. All four classifiers demonstrated high predictive performance, with accuracy scores exceeding 93%.

Having assessed each method individually, we now move on to **compare the models collectively** in terms of their predictive capabilities, robustness, and suitability for this particular dataset. This comparative analysis will help us identify the most effective classifier for the breast cancer diagnosis task.

Chapter 4

Comparative Analysis of Classification Models

In this section, we compare the performance of the four classification models applied to the breast cancer dataset:

Fisher’s Linear Discriminant Analysis (FDA), **Quadratic Discriminant Analysis (QDA)**, **Logistic Regression**, and **Probit Regression**. The comparison is based on standard performance metrics such as accuracy, precision, recall, F1-score, and cross-validation scores.

Table 4.1: Comparative Summary of Classification Models

Model	Accuracy	Precision	Recall	F1-score	CV Score (%)
FDA	0.956	0.96	0.96	0.96	95.61 ± 2.51
QDA	0.939	0.94	0.94	0.94	95.43 ± 2.51
Logistic Regression	0.956	0.96	0.96	0.96	93.32 ± 3.13
Probit Regression	0.956	0.96	0.96	0.96	95.61 ± 2.38

From Table 4.1, it is evident that all four models achieve high classification performance, with accuracy values exceeding 93%. FDA, Logistic Regression, and Probit Regression each achieve an accuracy of approximately 95.6%, while QDA slightly lags with 93.9%. All models show consistent performance in terms of precision, recall, and F1-score, suggesting that they are equally effective in handling the classification task.

The cross-validation scores also support the robustness of these models across different splits of the dataset. While FDA and Probit Regression achieve the highest cross-validation scores (95.61%), Logistic Regression and QDA closely follow with 95.43% and 93.32% respectively. This consistency reinforces the models’ reliability and generalizability.

In terms of interpretability and robustness to assumptions, FDA is theoretically appealing under the assumption of normally distributed features with equal covariance, whereas QDA is more flexible and can handle unequal covariances. Logistic and Probit regressions are regression-based models that do not require the assumption of normally distributed features and are both interpretable through their coefficients.

Overall, while all models perform similarly well, one might prefer Probit Regression or FDA for their slightly better cross-validation scores and robust performance. The choice between them can be further guided by interpretability needs or computational considerations.

4.1 ROC Curves

The **Receiver Operating Characteristic (ROC)** curve is a graphical representation of a classifier’s performance. It shows the trade-off between the *True Positive Rate (TPR)* or *Recall*, and the *False Positive Rate (FPR)* across various threshold values. The ROC curve is plotted with the FPR on the x-axis and the TPR on the y-axis, which provides insight into the performance of a classification model at different thresholds.

True Positive Rate (TPR), also known as recall, is the proportion of actual positives that are correctly identified by the model. It is calculated as:

$$TPR = \frac{TP}{TP + FN}$$

where TP is the number of true positives and FN is the number of false negatives.

False Positive Rate (FPR) is the proportion of actual negatives that are incorrectly classified as positives. It is calculated as:

$$FPR = \frac{FP}{FP + TN}$$

where FP is the number of false positives and TN is the number of true negatives.

The **Area Under the Curve (AUC)** is a scalar value that summarizes the performance of the model. AUC represents the probability that the classifier will rank a randomly chosen positive instance higher than a randomly chosen negative instance. The AUC value ranges from 0 to 1, with the following interpretation:

- AUC = 0.5: The model performs no better than random guessing.
- AUC > 0.5: The model performs better than random guessing, with a higher AUC indicating better performance.
- AUC = 1: The model perfectly classifies all instances without any errors.

A higher AUC indicates better overall performance of the classifier. In practical terms, an AUC value above 0.7 is often considered acceptable, while values above 0.9 typically indicate excellent performance. Comparing the ROC curves of multiple models allows us to visually assess and compare their performance at different classification thresholds.

In our case, the ROC curves for the four models (Fisher Discriminant Analysis, Quadratic Discriminant Analysis, Logistic Regression, and Probit Regression) will provide a comprehensive understanding of how each model distinguishes between benign and malignant cases in breast cancer detection. The AUC values will further highlight the most effective model for this task. Here is the figure including ROC Curves for the 4 different classification models we used -

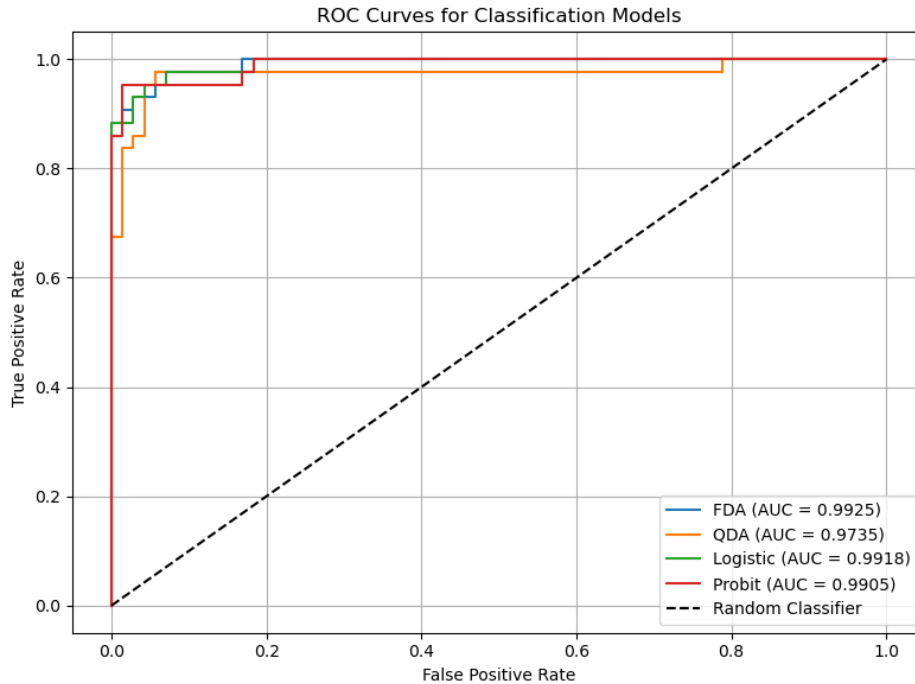


Figure 4.1: Comparison of ROC Curves for All Four Classification Models

Interpretation of ROC Curves

The ROC curves plotted for the four classification models—Fisher Discriminant Analysis (FDA), Quadratic Discriminant Analysis (QDA), Logistic Regression, and Probit Regression—offer a visual and quantitative comparison of their performance. The Area Under the Curve (AUC) values for these models are as follows: **FDA: 0.9925, QDA: 0.9735, Logistic Regression: 0.9918, and Probit Regression: 0.9905.**

These AUC values, all being significantly close to 1, indicate that all four models are highly effective at distinguishing between the benign and malignant cases. The FDA model exhibits the highest AUC, implying the best overall classification performance, followed very closely by Logistic and Probit regression models. QDA, while slightly behind the others, still performs excellently with an AUC of 0.9735.

From a visual standpoint, the ROC curves of all models hug the top-left corner of the plot, which signifies high true positive rates and low false positive rates across all threshold values. The close proximity of AUC values across the models suggests that all are well-suited for this classification task, although FDA shows a marginally superior discriminatory capability. This analysis confirms the robustness and reliability of the chosen models in classifying breast cancer cases with high accuracy.

Conclusion

In this project, we carried out a comprehensive analysis of the breast cancer dataset with the objective of accurately classifying tumors as benign or malignant. We began by preprocessing the data and addressing multicollinearity among the predictor variables to ensure model stability and interpretability. Subsequently, we performed a test for multivariate normality using the Kolmogorov-Smirnov test based on Mahalanobis distances, which guided our choice of appropriate classification models.

We applied four different classification techniques—Fisher Discriminant Analysis (FDA), Quadratic Discriminant Analysis (QDA), Logistic Regression, and Probit Regression—to the dataset. Each model was thoroughly evaluated using a range of performance metrics including accuracy, precision, recall, F1-score, confusion matrix, cross-validation scores, and ROC-AUC values.

Finally, we compared the models based on their classification performance. All four models performed exceptionally well, with FDA showing the highest AUC and slightly better overall metrics. This systematic approach not only demonstrated the power of classical statistical methods in classification tasks but also emphasized the importance of proper data preparation and model evaluation in deriving meaningful and reliable insights from real-world datasets.